

ECON 154: The Global Economy: Trade, Finance, and Policy

Lecture 4 Notes

Markets and Prices II

Firms in Markets: Costs and Supply

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Contents

Learning Objectives

After this lecture you should be able to:

1. Explain why firms exist and the role of specialization and coordination.
2. Distinguish revenue, economic cost, and profit.
3. Distinguish fixed, variable, and total costs in the short run.
4. Explain why firms make production decisions at the margin.
5. Define supply and quantity supplied and explain the law of supply.
6. Construct individual and market supply schedules and curves.
7. Distinguish a movement along a supply curve from a shift of supply.
8. Identify the major factors that shift supply.
9. Apply the supply framework to coffee, oil, and Maine lobster markets.

1 Introduction: Why Do Firms Exist?

In the previous lecture we studied consumers. We asked how individuals decide whether to purchase a good or service and introduced the concept of demand. Demand summarizes the quantities that consumers are willing and able to purchase at different prices. Behind every demand curve lies a series of individual decisions: should I buy one more coffee, one more concert ticket, or one more gallon of gasoline?

Markets, however, require another participant. Consumers can only purchase goods if someone first decides to produce them. Every cup of coffee, every smartphone, every loaf of bread, and every airplane ticket exists because a firm chose to organize resources and produce something that consumers value.

This lecture studies the producer's side of the market. Instead of asking

“Should I buy this?”

we ask

“Should I produce this?”

Although the perspective changes, the underlying economic reasoning is remarkably similar. Just as consumers compare benefits and costs before making a purchase, firms compare revenues and costs before deciding how much to produce.

Economic Intuition

Consumers maximize satisfaction subject to limited income.

Firms seek to earn profits by transforming scarce resources into products that consumers value.

Understanding how firms make decisions is essential because the producer side of the market determines the supply of goods and services. In the next lecture we will combine consumer demand with producer supply to explain how markets determine prices.

2 A Motivating Question

Consider the following question.

Why doesn't everyone simply make everything they consume?

Suppose you enjoy drinking coffee every morning.

Instead of purchasing coffee from a café, you could

- grow coffee trees,
- harvest the beans,
- roast them,
- grind them,
- build an espresso machine,
- make your own milk,
- bake your own pastry.

Almost nobody does this.

Instead, thousands of people around the world specialize in small tasks.

Coffee is grown in Brazil or Colombia.

Machines are designed in Italy.

Paper cups are manufactured elsewhere.

Transportation companies move the beans.

Baristas prepare the final product.

Consumers simply purchase the finished cup.

Why?

Economics provides a simple answer.

Definition

Specialization allows individuals and firms to concentrate on activities in which they are relatively productive.

Trade then allows everyone to consume far more than they could produce alone.

This idea will become central later in the semester when we study comparative advantage and international trade. For now, it motivates why firms exist in the first place.

3 The Role of Firms in the Economy

A firm is an organization that combines resources in order to produce goods or services for sale.

Resources include

- labor,
- machinery,
- buildings,
- land,
- technology,
- knowledge,
- financial capital.

The purpose of the firm is to transform these inputs into outputs that consumers value.

Figure ?? summarizes this idea.



Figure 1: The basic production process.

Notice that firms do not create value simply because they exist. They create value only if consumers are willing to purchase what they produce.

This observation immediately links the producer side of the market with the consumer side studied in Lecture 3.

4 Entrepreneurs and Firms

Many firms begin with an entrepreneur.

An entrepreneur is an individual who identifies an opportunity to create value by introducing a new product, improving an existing product, or producing a good more efficiently.

Examples include

- opening a neighborhood coffee shop,

- starting a landscaping business,
- developing a new software application,
- designing a more efficient manufacturing process.

Entrepreneurs take risks because there is no guarantee that consumers will purchase their product.

Some businesses succeed.

Many fail.

The possibility of earning profits encourages entrepreneurs to innovate, while the possibility of losses discourages inefficient production.

Real-World Example

A coffee shop owner must decide whether enough customers are willing to purchase coffee each morning to cover the costs of rent, labor, coffee beans, electricity, and equipment. If revenues exceed costs, the business earns a profit. If costs exceed revenues, the business incurs a loss.

Notice how similar this reasoning is to consumer choice.

Consumers compare benefits and costs before buying.

Firms compare revenues and costs before producing.

5 The Coffee Shop Running Example

Throughout the next several lectures we will use a small independent coffee shop as our primary example.

Although coffee may seem simple, it illustrates nearly every important concept in introductory economics.

The owner must answer questions such as

- Should the shop open on Sundays?
- Should another employee be hired?
- Should prices increase?
- Should another espresso machine be purchased?
- Should the shop stay open one additional hour?

Each of these decisions reflects the same underlying principle:

Economic decisions require comparing benefits and costs at the margin.

By following the same coffee shop throughout several lectures, we will see how demand, supply, market equilibrium, elasticity, taxation, and even international trade fit together into one coherent framework.

Ask an Economist

Ask an Economist

Student:

“If consumers ultimately decide what they buy, do firms really have any control over the market?”

Answer:

Consumers determine what they would like to purchase, but firms determine what they are willing to produce. Markets work because prices coordinate these two independent decisions. Understanding the producer’s decision is therefore just as important as understanding consumer demand.

Checkpoint: Why Firms Exist

1. Markets require both consumers and producers.
2. Firms exist because specialization increases productivity.
3. Firms combine inputs to produce goods and services.
4. Entrepreneurs organize production while accepting risk.
5. Producer decisions, like consumer decisions, compare benefits and costs.
6. Throughout this lecture we will study how firms decide how much to produce.

6 The Objective of the Firm

Every day, millions of firms around the world make production decisions.

A neighborhood bakery decides how many loaves of bread to bake before sunrise.

A software company decides whether to hire another programmer.

An airline decides whether to add another flight between Boston and Chicago.

A coffee shop decides whether it is worthwhile to stay open one additional hour.

Although these firms operate in very different industries, they all face the same basic economic problem:

Should we produce more?

The answer depends on comparing the additional benefits from production with the additional costs.

Consumers compare benefits and costs before making purchases.

Firms compare revenues and costs before producing goods and services.

Definition

A firm's objective is generally assumed to be the maximization of profit. Profit measures the difference between the revenues a firm earns from selling its products and the costs incurred in producing them.

$$\text{Profit} = \text{Total Revenue} - \text{Total Economic Cost}$$

Throughout this course we will typically assume firms maximize profit. This is a useful benchmark, although some organizations (such as nonprofits, cooperatives, or public enterprises) may pursue additional objectives.

Notice that economists do not assume firms seek to maximize sales or revenue. A business that sells many products while consistently losing money will eventually be forced to close. Profit provides the incentive for firms to remain in business, invest, innovate, and expand.

7 Revenue

Revenue represents the money a firm receives from selling its products.

Suppose a coffee shop sells 180 cups of coffee during a morning at a price of \$5 per cup.

The total revenue earned is

$$\text{Revenue} = \$5 \times 180 = \$900.$$

More generally,

$$\text{Revenue} = \text{Price} \times \text{Quantity Sold}$$

Revenue therefore depends on two factors.

- How many units are sold.
- The price received for each unit.

Revenue is often relatively easy to observe because firms record every sale. Costs, however, are frequently more difficult to measure.

Real-World Example

Imagine that Starbucks increases the price of a latte from \$5.50 to \$6.00. Revenue could increase because each coffee sells for a higher price. However, if enough customers decide not to buy coffee, revenue could also decrease. Revenue therefore depends on both price and consumer demand.

8 Understanding Costs

Although revenue is important, firms cannot focus only on the money they receive.

Producing goods requires resources, and those resources are scarce.

Coffee beans must be purchased.

Employees must be paid.

Buildings must be rented.

Equipment eventually wears out.

Economists refer to these sacrifices collectively as production costs.

Definition

Production costs are the value of the resources used in producing a good or service.

An important lesson in economics is that costs are measured in terms of opportunity cost rather than simply accounting expenditures.

Suppose the owner of a coffee shop works behind the counter without paying herself a salary.

From an accounting perspective, labor costs appear lower.

From an economic perspective, however, the owner's time still has value because she could have worked elsewhere.

Economists therefore include the opportunity cost of resources, even when no explicit payment is made.

Economic Intuition

Every productive resource has an opportunity cost.

Even if no money changes hands, using a resource for one purpose means giving up the opportunity to use it elsewhere.

9 Fixed and Variable Costs

Not every production cost behaves in the same way.

Some expenses must be paid regardless of how much the firm produces.

Other expenses increase every time another unit is produced.

Understanding this distinction is essential because it explains why firms respond differently to changes in demand and prices.

Fixed Costs

Fixed costs are expenses that do not depend on current production.

Whether the coffee shop serves zero customers or three hundred customers today, these expenses must still be paid.

Typical examples include

- monthly rent,
- business insurance,
- software subscriptions,
- property taxes,
- some equipment leases.

These costs remain approximately constant over short periods of time.

Economic Intuition

Whether a cost is fixed depends on the time horizon. Rent may be fixed over the duration of a lease but adjustable once the lease expires. Throughout this lecture we focus on the short run.

Variable Costs

Variable costs change with production.

Every additional latte requires

- coffee beans,
- milk,
- paper cups,
- electricity,
- labor time.

If the coffee shop serves twice as many customers, these costs generally increase.

Total Costs

Total cost combines both components.

$$\text{Total Cost} = \text{Fixed Cost} + \text{Variable Cost}$$

Figure ?? summarizes the relationship.



Figure 2: Total cost equals fixed costs plus variable costs.

10 The Coffee Shop Example

Throughout the next several lectures we will continue to follow the same independent coffee shop.

Suppose the owner estimates the following daily costs.

Item	Type	Daily Cost
Rent	Fixed	\$120
Insurance	Fixed	\$25
Coffee machine lease	Fixed	\$35
Coffee beans	Variable	Depends on sales
Milk	Variable	Depends on sales
Paper cups	Variable	Depends on sales
Barista wages	Variable/Step Cost	Depends on staffing decisions

Notice that if no customers arrive today, the owner still pays the rent, insurance, and equipment lease.

However, almost no coffee beans or milk are used.

Variable costs remain very low.

Now suppose demand unexpectedly doubles.

The coffee shop must purchase more beans, more milk, more cups, and perhaps schedule another employee.

Variable costs increase while fixed costs remain unchanged.

This distinction explains why firms often react gradually to increases in demand.

Common Mistake

Students often believe firms maximize revenue.

They do not.

A business earning \$50,000 in revenue but spending \$60,000 on production loses money.

Economists therefore assume firms seek to maximize profit, not revenue.

Ask an Economist**Student:**

“Why wouldn’t a coffee shop simply sell as many coffees as possible?”

Economist:

Because producing additional coffees also requires additional resources.

As production expands, costs increase.

The relevant question is therefore not

How many coffees can we sell?

but rather

Is producing one more coffee worth its additional cost?

This question leads directly to the next concept:

Marginal Cost

which becomes the foundation of the supply curve.

Checkpoint: Revenue and Costs

1. Firms seek to maximize profit rather than revenue.
2. Revenue equals price multiplied by quantity sold.
3. Economic costs include opportunity costs.
4. Fixed costs do not depend on current production.
5. Variable costs increase as output increases.
6. Total cost equals fixed cost plus variable cost.
7. The next step is to understand how costs change as firms produce one additional unit.

11 Marginal Analysis and the Firm’s Production Decision

Imagine that it is 2:30 in the afternoon. The morning rush is over, but customers still arrive occasionally. A customer orders one more latte. Should the coffee shop prepare it?

The rent, insurance, and espresso-machine lease have already been paid for the day. Those fixed costs matter for the overall profitability of the business, but they do not change because of this one order. The immediate decision is narrower:

What additional revenue will the latte generate, and what additional cost will producing it require?

Economists call this **marginal analysis**. The word marginal means additional or incremental.

Definition

Marginal cost is the additional cost incurred by producing one more unit of output.

For one additional latte, the relevant costs may include coffee beans, milk, a cup, electricity, and a few minutes of labor. The daily rent is not part of the marginal cost because it does not change when this latte is produced.

11.1 Why Marginal Cost Often Rises

When production is low, the shop may have spare capacity. The barista is not rushed, the espresso machine is available, and producing another latte is relatively easy.

During the busiest part of the morning, however, another latte may require employees to work faster, delay other orders, use overtime, or call in another worker. As the shop approaches its current capacity, additional production becomes more costly.

Economic Intuition

As output expands in the short run, firms eventually rely more heavily on scarce or less productive resources. This tends to make the marginal cost of additional output rise.

Table ?? provides an illustrative example.

Table 1: Illustrative Marginal Costs for a Coffee Shop

Daily Latte Output	Total Variable Cost	Marginal Cost of Next Block
0	\$0	–
20	\$42	\$2.10 per latte
40	\$88	\$2.30 per latte
60	\$140	\$2.60 per latte
80	\$202	\$3.10 per latte
100	\$280	\$3.90 per latte
120	\$382	\$5.10 per latte

The numbers are hypothetical. Their purpose is to show the logic: additional output becomes progressively more expensive as the firm stretches its short-run capacity.

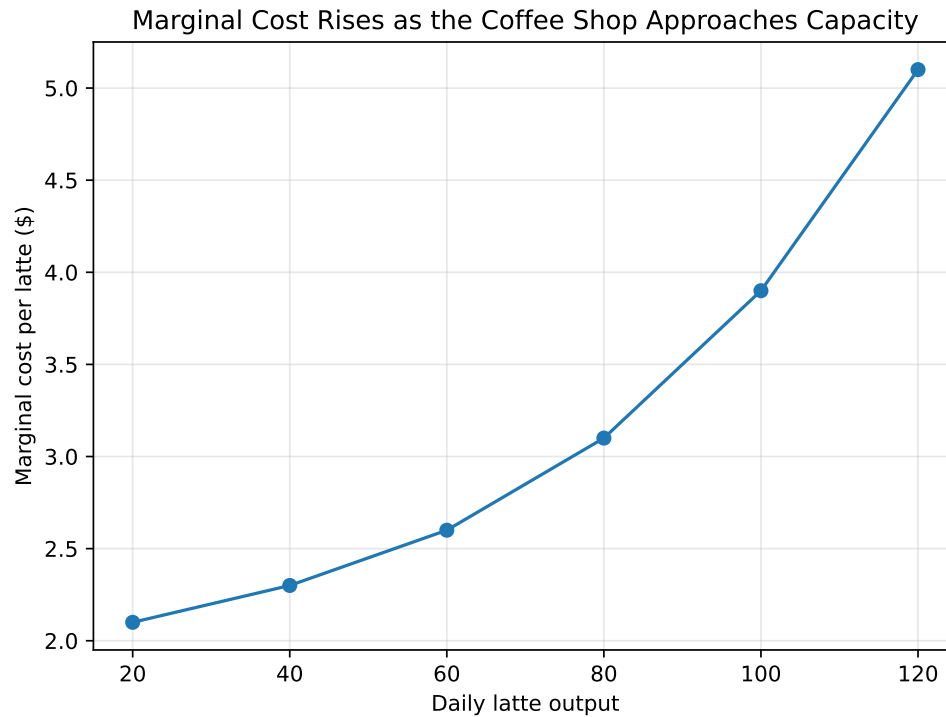


Figure 3: An illustrative marginal-cost curve for the coffee shop.

12 Supply and Quantity Supplied

Supply describes the producer side of a market.

Definition

Supply is the complete relationship between the price of a good and the quantities producers are willing and able to offer for sale at those prices.

Quantity supplied is the particular quantity producers wish to sell at one specific price.

This distinction exactly parallels the distinction between demand and quantity demanded from Lecture 3.

Common Mistake

A change in the good's own price normally changes **quantity supplied** and produces a movement along the existing supply curve. It does not shift supply.

12.1 The Law of Supply

Holding other relevant factors constant, a higher price generally leads firms to offer a larger quantity for sale, while a lower price leads them to offer a smaller quantity.

$$P \uparrow \implies Q_s \uparrow$$

$$P \downarrow \implies Q_s \downarrow$$

Higher prices do not make firms suddenly more interested in profit. Firms already prefer profit. Instead, a higher price makes additional units profitable even when those units have relatively high marginal costs.

Economic Intuition

A firm produces an additional unit when the revenue received from that unit is at least as large as its marginal cost. As price rises, more high-cost units pass this profitability test.

13 Supply Schedules and Supply Curves

A supply schedule is a table showing the quantity a producer is willing and able to sell at different prices.

Table 2: Illustrative Supply Schedule for the Coffee Shop

Price per Latte	Quantity Supplied per Day
\$2	0
\$3	40
\$4	80
\$5	100
\$6	120
\$7	135

A supply curve displays the same information graphically. Economists place price on the vertical axis and quantity on the horizontal axis, just as they do for demand.

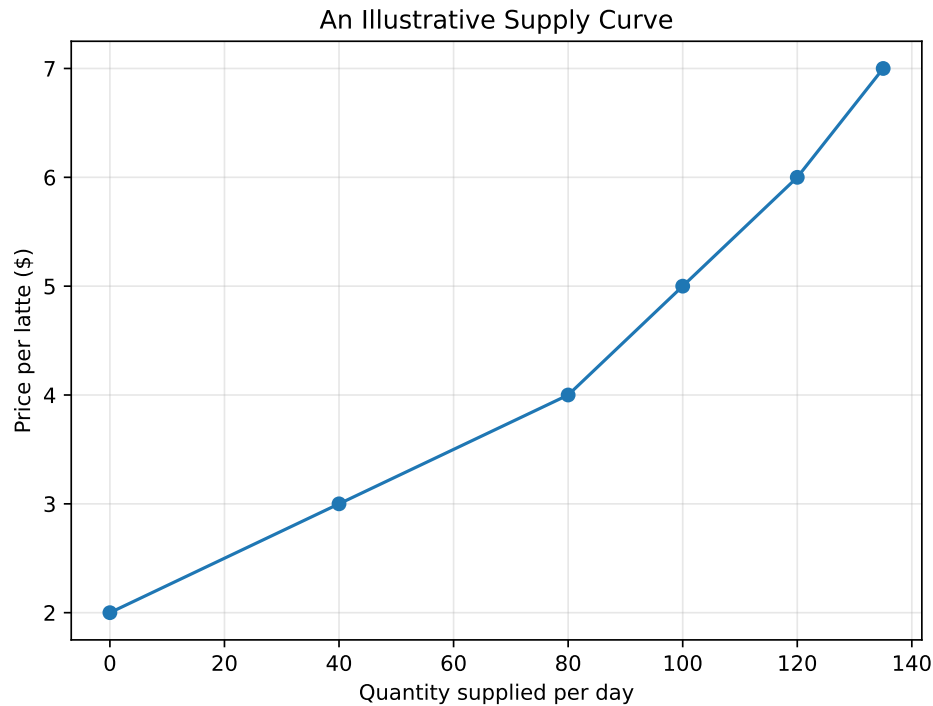


Figure 4: An illustrative upward-sloping supply curve.

The curve slopes upward because higher prices make progressively more costly units profitable to produce.

14 Individual Supply and Market Supply

An individual supply curve describes one firm. Market supply describes the total quantity all firms are willing and able to sell at each price.

To construct market supply, we add the quantities supplied by individual firms at the same price. This is called a horizontal sum.

Table 3: From Individual to Market Supply

Price	Firm A	Firm B	Firm C	Market Quantity Supplied
\$2	0	0	0	0
\$3	20	10	10	40
\$4	40	30	20	90
\$5	55	45	35	135
\$6	65	60	50	175
\$7	75	70	65	210

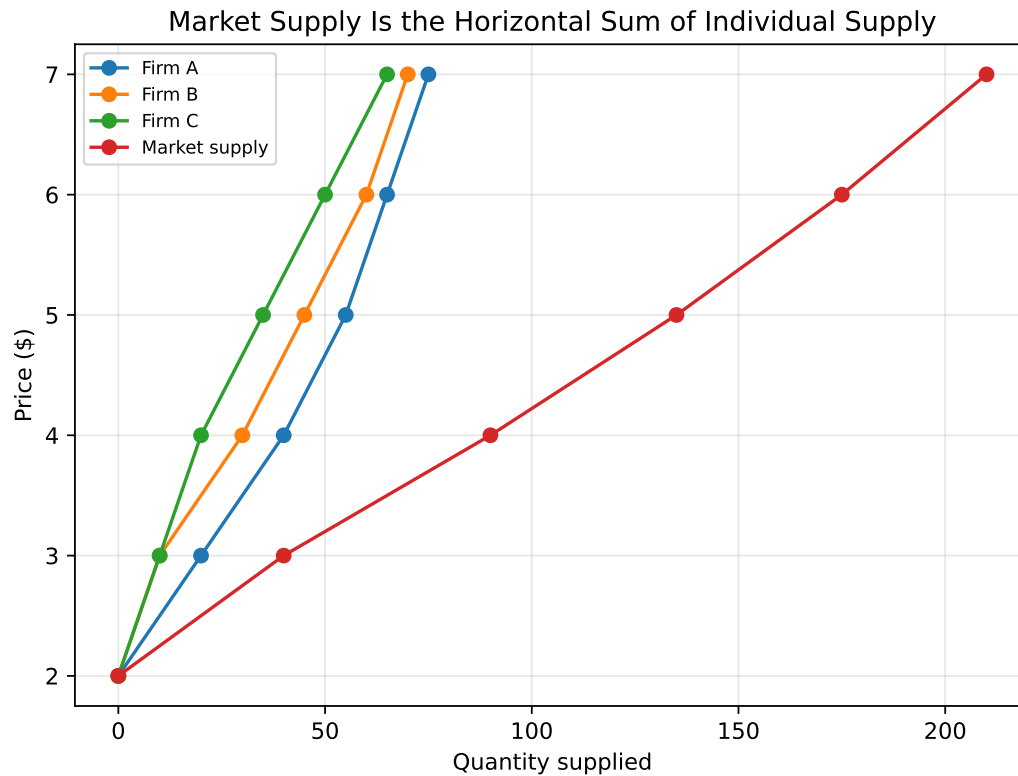


Figure 5: Market supply is the horizontal sum of individual firms' supply curves.

Market supply can increase because existing firms expand production or because additional firms enter the market. It can decrease because firms cut production or leave the market.

15 Movements Along Supply and Shifts of Supply

When the good's own price changes and other relevant factors remain constant, quantity supplied changes. This appears as a movement along the supply curve.

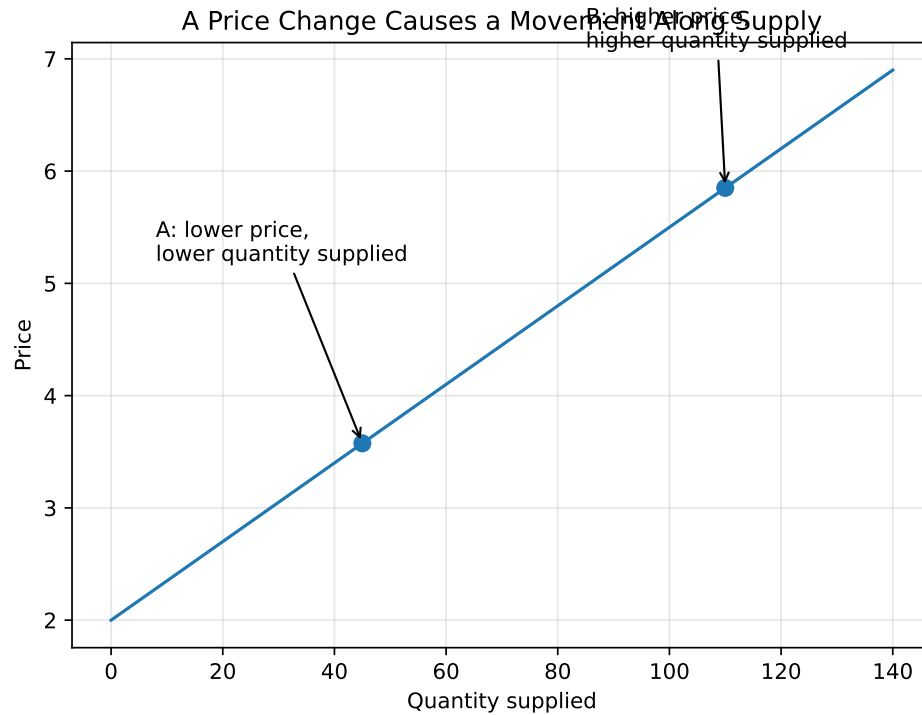


Figure 6: A change in the good's own price causes a movement along the supply curve.

Supply shifts when a factor other than the good's own price changes the quantity firms wish to sell at every possible price.

An increase in supply shifts the curve to the right: firms offer more at each price. A decrease in supply shifts it to the left.

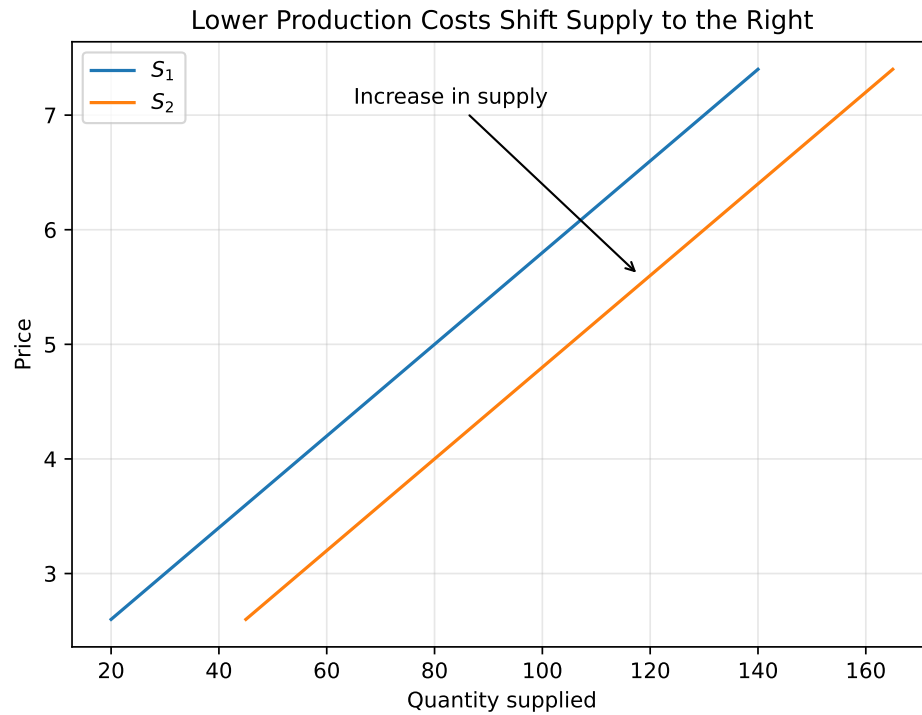


Figure 7: A reduction in production costs increases supply and shifts the curve to the right.

16 What Shifts Supply?

The main supply shifters are summarized in Figure ??.

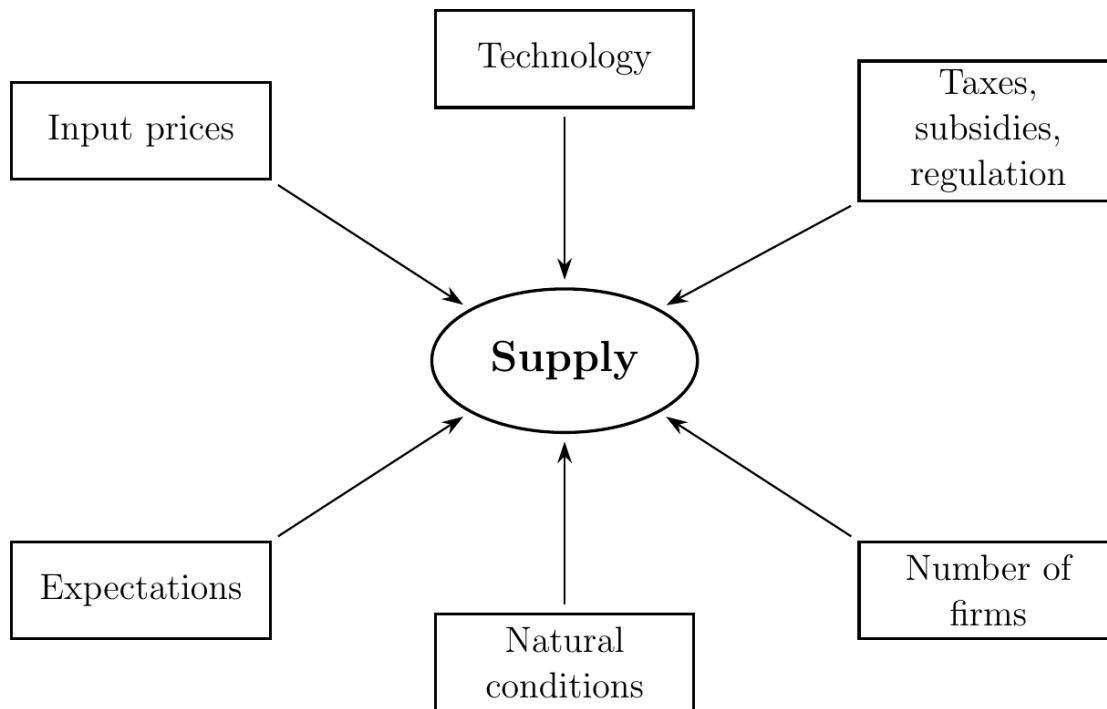


Figure 8: Major factors that shift supply.

16.1 Input Prices

A fall in the price of coffee beans, milk, electricity, or labor reduces production costs. The coffee shop can profitably offer more lattes at each price, so supply shifts to the right.

Higher input prices increase costs and shift supply to the left.

16.2 Technology and Productivity

Improved equipment, better organization, or worker training can allow a firm to produce more output with the same resources. This increases supply.

Technology need not be a sophisticated invention. A more efficient ordering system or a better shop layout may also reduce costs.

16.3 Taxes, Subsidies, and Regulation

A tax imposed on production raises the cost of supplying the good and tends to reduce supply. A subsidy lowers the producer's net cost and tends to increase supply.

Regulations may raise costs when firms must purchase equipment or change production methods.

Regulations can also create benefits not reflected in the firm's private costs, such as safer workplaces or cleaner air. Whether a regulation is desirable is a separate normative question from how it shifts supply.

16.4 Expectations

If sellers expect a storable product's price to rise significantly in the near future, they may withhold some output today. Current supply falls.

Expectations operate differently for products that cannot be stored easily. A café cannot save today's prepared latte for sale next month.

16.5 Natural Conditions

Weather, disease, fires, floods, and biological conditions are especially important for agriculture, fishing, energy, and transportation.

Good growing conditions may increase crop supply. Drought or disease may reduce it.

16.6 Number of Firms

When firms enter a market, market supply generally increases. When firms exit, market supply decreases.

Common Mistake

"More firms entered because the product's price rose" describes two connected steps. The higher price first creates a movement along existing firms' supply curves. If the higher profitability later attracts entry, the number of firms rises and market supply shifts to the right.

17 Economics in the News: Coffee Supply

Coffee is an internationally traded agricultural product. Its supply can be affected by drought, frost, disease, labor conditions, shipping costs, and exchange rates.

Suppose drought damages coffee plants in a major producing country. At any given market price, growers can bring fewer beans to market. The supply of coffee shifts left.

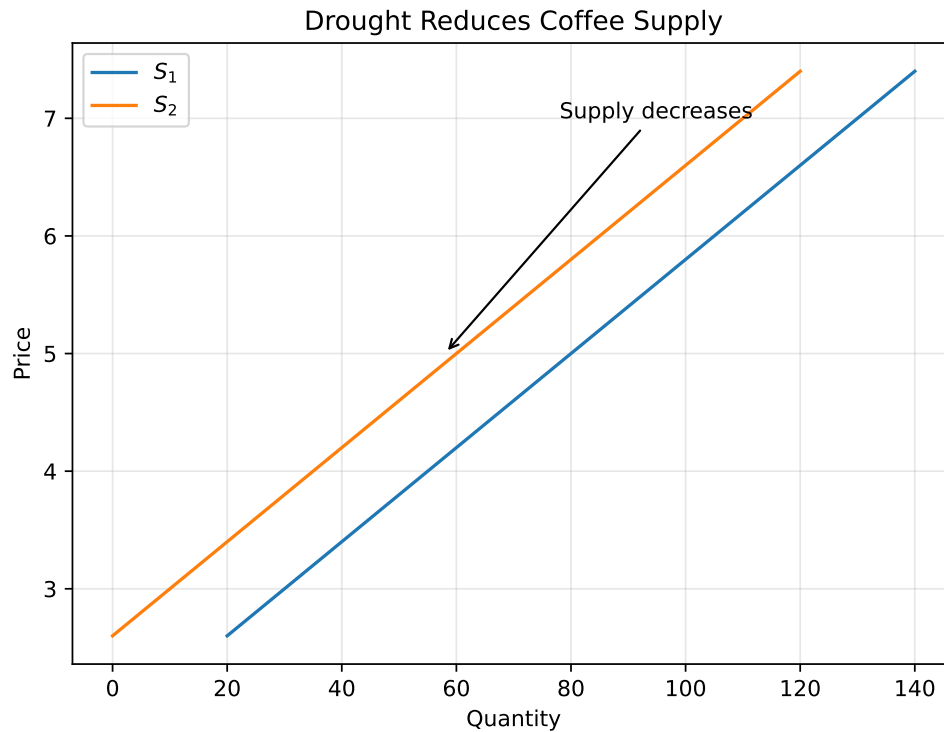


Figure 9: A negative production shock reduces coffee supply. The effect on the market price will be determined after supply is combined with demand in Lecture 5.

Economics in the News

A news article that reports “drought pushes coffee prices higher” contains two separate parts of an economic argument. Drought is a supply shifter. The prediction about price requires both supply and demand. We will complete that analysis in the next lecture.

18 Economics in the News: Oil and OPEC

Crude oil supply is global. Producers outside OPEC generally make independent investment and production decisions, while OPEC members coordinate production targets. The U.S. Energy Information Administration explains that reductions in OPEC production targets have historically tended to place upward pressure on crude-oil prices.

From the perspective of this lecture, a coordinated production cut reduces the quantity of oil offered to the world market at each relevant price. It is therefore represented as a leftward shift of oil supply.

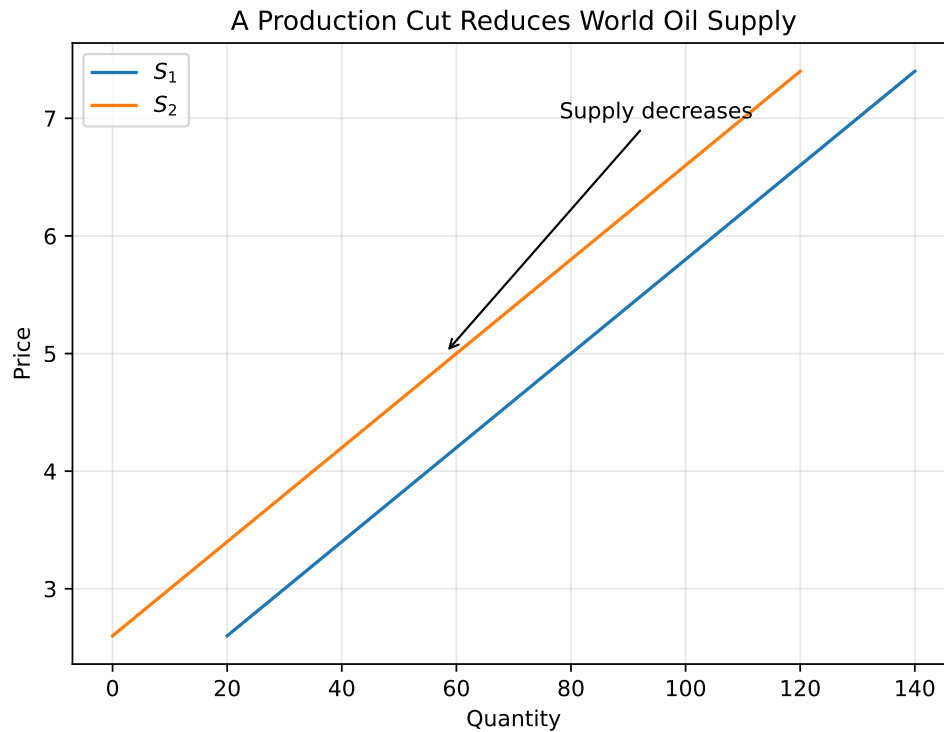


Figure 10: A coordinated production cut shifts the world supply of oil to the left.

The full price effect depends on the demand for oil and on the response of other producers. If non-OPEC producers expand output, they partly offset the initial reduction in supply.

19 Local Application: Maine Lobster

The Maine lobster industry offers a useful local example of supply.

Lobster supply depends on biological conditions, weather, fuel prices, labor availability, trap limits, minimum and maximum size rules, closed areas, and gear requirements. NOAA Fisheries divides the fishery into management areas and applies restrictions that differ across locations.

A severe storm may prevent boats from leaving port, reducing the quantity brought to market temporarily. A change in trap limits or a closure can reduce supply over a longer period. Improved equipment might increase the quantity that can be harvested with the same labor and fuel, although biological and regulatory constraints remain important.

Factors Affecting Maine Lobster Supply

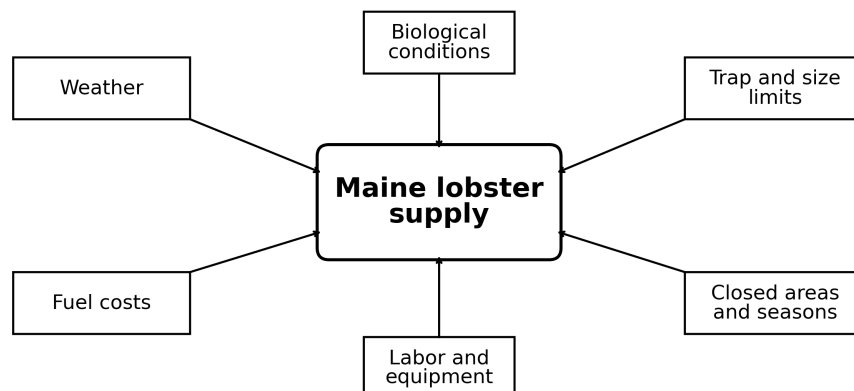


Figure 11: Examples of factors affecting the supply of Maine lobster.

Economic Intuition

Supply analysis does not say that environmental or fishery regulations are good or bad. It identifies one consequence: restrictions that reduce the quantity producers can legally or sustainably bring to market shift supply. Evaluation of the policy must also consider ecological and social benefits.

20 Common Misconceptions

Common Mistake

Supply is not quantity supplied.

Supply is the entire price-quantity relationship. Quantity supplied is one amount at one particular price.

Common Mistake

A higher price does not shift the supply curve.

It causes a movement along the curve. Input costs, technology, policy, expectations, natural conditions, or the number of firms shift supply.

Common Mistake

Firms do not maximize revenue.

A firm can have high sales and still lose money. The benchmark objective is profit: revenue minus economic cost.

Common Mistake**Fixed does not mean permanent.**

A cost can be fixed over today's production decision but adjustable over a longer horizon.

Common Mistake**Economic costs are not limited to cash payments.**

The owner's time, privately owned space, and invested funds all have opportunity costs.

21 A Practical Method for Reading Supply Stories

When reading a news report about producers, ask:

1. What is the market and what is the unit of output?
2. What price does the producer receive?
3. Is the story about a change in the good's own price or another production condition?
4. Is the result a movement along supply or a shift of supply?
5. Which cost, technology, expectation, policy, or natural condition changed?
6. Is the analysis focused on the short run or the long run?
7. Could other firms enter, exit, or offset the initial change?
8. What additional information about demand would be needed to predict the new market price?

Key Takeaways

1. Firms organize specialized resources to produce goods and services.
2. The standard benchmark is that firms seek to maximize profit, not revenue.
3. Economic cost includes explicit payments and opportunity costs.
4. Fixed costs do not vary with current output in the short run; variable costs do.
5. Marginal cost is the additional cost of producing one more unit.
6. Higher prices generally increase quantity supplied because they make additional high-cost units profitable.
7. Supply describes the complete relationship between price and quantity supplied.

8. Market supply is the horizontal sum of individual firms' quantities supplied.
9. A change in own price causes a movement along supply; changes in production conditions shift supply.
10. Input prices, technology, taxes and subsidies, regulation, expectations, natural conditions, and the number of firms can shift supply.

Looking Ahead: Markets in Action

Consumers have plans describing how much they wish to buy at different prices.

Firms have plans describing how much they wish to sell at different prices.

These plans need not initially agree.

How do prices coordinate buyers and sellers?

Lecture 5 combines demand and supply to explain market equilibrium, shortages, surpluses, and price adjustment. Once the equilibrium price is determined, we will return to the consumer benefit preview from Lecture 3 and introduce consumer surplus, producer surplus, and total gains from exchange together.

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